directions, for definiteness, are here called UD (for "up/down"), NS (for "north/south"), and EW (for "east/west"), and each of the three curves describes the behavior of the tidal force along one of these directions. Time is plotted horizontally. At any time when the UD curve is *above* the horizontal time axis, the tidal force is *stretching* along the UD direction, while at a time when the UD curve is *below* the axis, the un tidal force is *squeezing.* The higher the curve above the axis, the stronger the stretch; the lower the curve below the axis, the stronger the squeeze. Notice the following: (i) At any moment of time there is a squeeze along two directions and a stretch along one. (H) The tidal forces oscillate between stretch and squeeze; each oscillation is called a "cycle." (iii) The cycles are collected into "eras." During each era, one of the three directions is subjected to a fairly steady squeeze, while the other two oscillate between stretch and squeeze. (iv) When the era changes, there is a change of the steady direction. (v) As the singularity is approached, the oscillations become infinitely rapid and the tidal forces become infinitely strong. The details of the division of cycles into eras and the change of oscillation patterns at the beginning of each era are governed by what is sometimes called a "chaotic map."

Einstein's equation predicts that, as the astronaut reaches the singularity, the tidal forces grow infinitely strong, and their chaotic oscillations become infinitely rapid. The astronaut dies and the atoms from which his body is made become infinitely and chaotically distorted and mixed—and then, at the moment when everything becomes infinite (the tidal strengths, the oscillation frequencies, the distortions, and the mixing), spacetime ceases to exist.

The laws of quantum mechanics object. They forbid the infinities. Very near the singularity, as best we understand it in 1993, the laws of quantum mechanics merge with Einstein's general relativistic laws and completely change the "rules of the game." The new rules are called *quantum gravity.*

The astronaut is already dead, his body parts are already thoroughly mixed, and the atoms of which he was made are already distorted beyond recognition when quantum gravity takes over. But nothing is infinite. The "game" goes on.

Just when does quantum gravity take over, and what does it do? As best we understand it in 1993 (and our understanding is rather poor), quantum gravity takes over when the oscillating tidal gravity (spacetime curvature) becomes so large that it completely deforms all objects in about 10^{-43} second or less.[2] Quantum gravity then radically changes the character of spacetime: It ruptures the unification of space and time into spacetime. It unglues space and time from each other, and then destroys time as a concept and destroys the definiteness of space. Time ceases to exist; no longer can